

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

SEATBELT RETRACTOR AND PRETENSIONER (SRP) - LEFT/RIGHT

WARNINGS:

- TO AVOID ACCIDENTAL DEPLOYMENT AND POSSIBLE PERSONAL INJURY, THE BACKUP POWER SUPPLY MUST BE DEPLETED BEFORE REPAIRING OR REPLACING ANY AIR BAG SUPPLEMENTAL RESTRAINT SYSTEM (SRS) COMPONENTS. TO DEplete THE BACKUP POWER SUPPLY ENERGY, DISCONNECT THE BATTERY GROUND CABLE AND WAIT ONE MINUTE. FAILURE TO FOLLOW THIS INSTRUCTION MAY RESULT IN PERSONAL INJURY.
- Do not use a multimeter to probe the restraints control module. It is possible for the power from the meter battery to trigger the activation of the airbags. Failure to follow this instruction may result in personal injury.

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding

bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

- It is advisable not to use a cellular phone or to have a cellular phone in close proximity when working on the restraints control module or associated systems.
- Given the legal implications of a restraints system failure, harness repairs to Air Bag module circuits are not acceptable. Where the text refers to "REPAIR the circuit", this will normally mean the replacement of a harness.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Seatbelt Retractor and Pretensioner (SRP) - Left/Right. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Seatbelt System](#) (501-20A Seatbelt System, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	▪ Invalid data received from another control module via the high speed CAN bus (chassis)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-86	High Speed CAN Communication Bus - Signal invalid	▪ Invalid signal received from another control module via the high speed CAN bus (chassis)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid signal source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-87	High Speed CAN Communication Bus - Missing	▪ Missing message from another control module	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the

	message	via the high speed CAN bus (chassis)	missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-88	High Speed CAN Communication Bus - Bus off	High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-00	Internal Control Module Software Incompatibility - No sub type information	- Master car configuration file not being transmitted by body control module / gateway module assembly - Car configuration file mismatch with vehicle specification - Driver/passenger seatbelt retractor and pretensioner is not configured correctly - Incorrect driver/passenger seatbelt retractor and pretensioner installed	- Using the manufacturer approved diagnostic system, check the body control module / gateway module assembly for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary - Using the manufacturer approved diagnostic system, re-configure the driver/passenger seatbelt retractor and pretensioner with the latest level software - Install a new driver/passenger seatbelt retractor and pretensioner as necessary
U2001-68	Reduced System Function - Event information	Driver/passenger seatbelt retractor and pretensioner has exceeded the design limit for durability	Confirm that the driver/passenger seatbelt retractor and pretensioner has exceeded the design limit for durability: Using the manufacturer approved diagnostic system, check

			datalogger signal - Dynamic Seatbelt Health Status (0x5824). If the result is 'not healthy' (0x00), install a new driver/passenger seatbelt retractor and pretensioner as necessary. If the result is 'healthy' (0x01), clear the DTCs and perform routing - Test Dynamic Seatbelt Retract and Release (0x5005)
U201A-57	Control Module Main Calibration Data - Invalid/incomplete software component	Driver/passenger seatbelt retractor and pretensioner is not configured correctly	Using the manufacturer approved diagnostic system, re-configure the driver/passenger seatbelt retractor and pretensioner with the latest level software
U2100-00	Initial Configuration Not Complete - No sub type information	Car configuration file mismatch with vehicle specification	Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary
U2101-00	Control Module Configuration Incompatible - No sub type information	Car configuration file mismatch with vehicle specification	NOTE: A driver/passenger seatbelt retractor and pretensioner cannot be installed in another vehicle as it has the original VIN permanently programmed into it Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and perform routines - On Demand Self Test (0x0202) - Test Dynamic Seatbelt Retract and Release (0x5005) - Clear Self-learning Data (0x0402)
U2300-54	Central Configuration - Missing calibration	Car configuration file mismatch with vehicle	Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary

		specification	
U3000-45	Control Module - Program memory failure	▪ Driver/passenger seatbelt retractor and pretensioner internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routines - On Demand Self Test (0x0202) - Test Dynamic Seatbelt Retract and Release (0x5005). If the fault persists, install a new driver/passenger seatbelt retractor and pretensioner
U3000-49	Control Module - Internal electronic failure	▪ Driver/passenger seatbelt retractor and pretensioner internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routines - On Demand Self Test (0x0202) - Test Dynamic Seatbelt Retract and Release (0x5005). If the fault persists, install a new driver/passenger seatbelt retractor and pretensioner
U3000-4B	Control Module - Over temperature	▪ Driver/passenger seatbelt retractor and pretensioner motor temperature too high	▪ Allow the driver/passenger seatbelt retractor and pretensioner motor to cool. Using the manufacturer approved diagnostic system, clear the DTCs and perform routines - On Demand Self Test (0x0202) - Test Dynamic Seatbelt Retract and Release (0x5005). If the fault persists, install a new driver/passenger seatbelt retractor and pretensioner
U3000-92	Control Module - Performance or incorrect operation	▪ Driver/passenger seatbelt retractor and pretensioner maximum activation levels exceeded	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new driver/passenger seatbelt retractor and pretensioner
U3002-81	Vehicle Identification Number - Invalid serial data received	▪ Driver/passenger seatbelt retractor and pretensioner is not configured correctly	▪ Using the manufacturer approved diagnostic system, re-configure the driver/passenger seatbelt retractor and pretensioner with the latest level software

U3003-62	Battery Voltage - Signal compare failure	▪ Mismatch between the voltage at the driver/passenger seatbelt retractor and pretensioner and the voltage value broadcast on the CAN bus ▪ Driver/passenger seatbelt retractor and pretensioner power or ground circuit open circuit, high resistance	▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02) - and compare it to battery voltage ▪ Refer to the electrical circuit diagrams and check the driver/passenger seatbelt retractor and pretensioner power and ground circuits for open circuit, high resistance
U3006-16	Control Module Input Power "A" - Circuit voltage below threshold	▪ Driver/passenger seatbelt retractor and pretensioner power or ground circuit open circuit, high resistance ▪ Battery/charging system fault ▪ Driver/passenger seatbelt retractor and pretensioner internal failure	NOTE: This DTC is set when the driver/passenger seatbelt retractor and pretensioner voltage falls below 9V ▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and check the driver/passenger seatbelt retractor and pretensioner power and ground circuits for open circuit, high resistance ▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routines - On Demand Self Test (0x0202) - Test Dynamic Seatbelt Retract and Release (0x5005). If the fault persists, install a new driver/passenger seatbelt

			retractor and pretensioner
U3006-17	Control Module Input Power "A" - Circuit voltage above threshold	▪ Battery/charging system fault ▪ Driver/passenger seatbelt retractor and pretensioner internal failure	NOTE: This DTC is set when the driver/passenger seatbelt retractor and pretensioner voltage rises above 16V ▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Using the manufacturer approved diagnostic system, clear the DTCs and perform routines - On Demand Self Test (0x0202) - Test Dynamic Seatbelt Retract and Release (0x5005). If the fault persists, install a new driver/passenger seatbelt retractor and pretensioner